RNA Velocity in Embryonic Mouse Pancreas: scVelo Stochastic vs Dynamical

Single-cell splicing kinetics on the Bastidas-Ponce 2019 endocrine atlas

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Summary

We re-analyzed the canonical scVelo demonstration dataset, the Bastidas-Ponce et al. 2019 mouse embryonic pancreas (E15.5, 3,696 cells across 8 annotated lineages), benchmarking the two main RNA velocity inference modes in scVelo: **stochastic** (first- and second-order moments, fast direction estimate) and **dynamical** (full per-gene splicing kinetics with global latent time). The stochastic model fits all 7,197 expressed genes with mean per-cell confidence 0.79; the dynamical model fits a more conservative subset of 2,722 genes (38% of input) and yields cleaner geometry and higher confidence (0.86). Both modes recover the textbook developmental flow Ductal -> Ngn3-low EP -> Ngn3-high EP -> Pre-endocrine -> {Alpha, Beta, Delta, Epsilon}, with the dynamical mode adding a coherent latent-time gradient and a per-gene driver ranking that highlights canonical endocrine maturation markers (Pcsk2, Pak3, Tmem163).

Dataset

The data are bundled with scVelo and originate from Bastidas-Ponce et al. (*Development*, 2019). After standard scVelo preprocessing (`filter_and_normalize` with `min_shared_counts=20`, followed by moments computation on a 30-PC, 30-neighbor graph), the working object contains:

- **3,696 cells** across 8 author-annotated lineages: Ductal, Ngn3 low EP, Ngn3 high EP, Pre-endocrine, and the four mature endocrine fates (Alpha, Beta, Delta, Epsilon).
- **7,197 genes** retained, each with both spliced and unspliced counts above the shared-count threshold.

Methods

Stochastic velocity. Per-gene gamma estimated by generalized least squares on first- and second-order spliced/unspliced moments. Implemented as `scv.tl.velocity(mode='stochastic')`. This mode fits every gene that passes preprocessing and produces no NaN velocities.

Dynamical velocity. Per-gene transcription, splicing, and degradation rates recovered by likelihood maximization (`scv.tl.recover_dynamics`, n_jobs=8), followed by `scv.tl.velocity(mode='dynamical')`. Genes whose kinetics cannot be reliably fit are returned as NaN. A global latent time is then inferred via `scv.tl.latent_time`.

Cluster-level summary. PAGA (`scv.tl.paga`) was applied to the velocity graph of the dynamical model, yielding a directed cluster-transition matrix and a graph plot overlaid on the UMAP.

Quantitative comparison. For each cell we computed the cosine similarity between its stochastic and dynamical velocity vectors, restricted to the 2,722 genes finite in both models. Per-cluster summaries were then aggregated.

Results

Velocity streams (Figure 1)

Both modes recover the canonical pancreatic endocrine flow on the UMAP: Ductal cells (top right) feed into Ngn3-low and then Ngn3-high endocrine progenitors, which converge into the Pre-endocrine population and fan out into Alpha, Beta, Delta, and Epsilon. Visually the directional fields are highly similar; the dynamical streamlines are smoother because per-gene kinetics regularize noisy genes out.

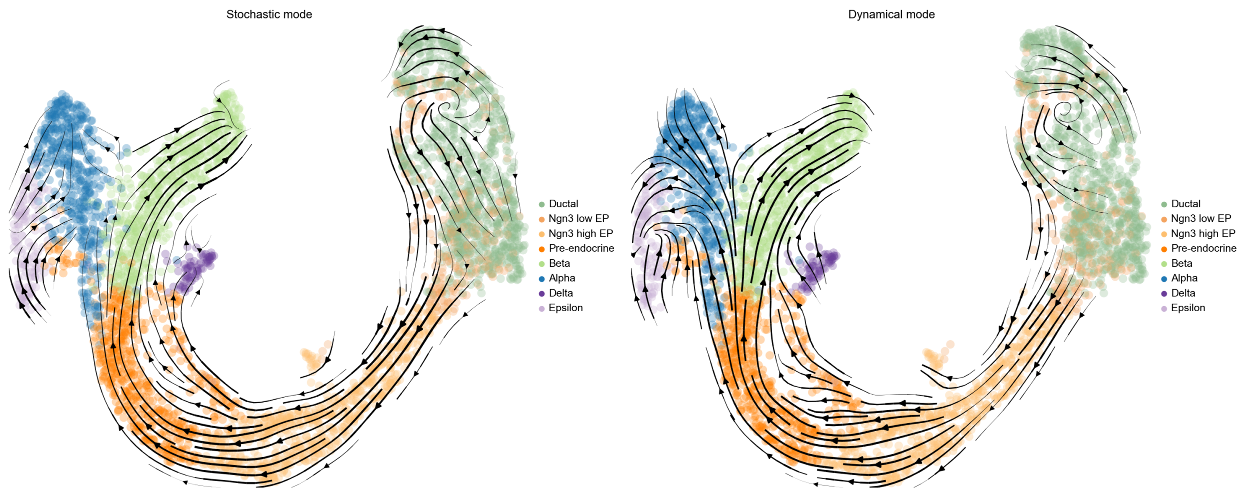

Figure 1: Side-by-side velocity streams. Left: stochastic mode; right: dynamical mode. Both recover the same lineage flow Ductal -> Ngn3 EP -> Pre-endocrine -> fate decisions. Streamlines are smoother in the dynamical mode.

Velocity length and confidence (Figure 2)

Velocity *length* (vector magnitude) is highest in the Pre-endocrine to Beta transition - the region where splicing kinetics is strongest, consistent with the massive ramp-up of insulin and secretory-granule transcription as cells mature. Velocity *confidence* (per-cell coherence with neighbors) is high across the entire trajectory except in pockets of Ductal cells, where a mixture of cycling and quiescent stem cells flattens the local flow.

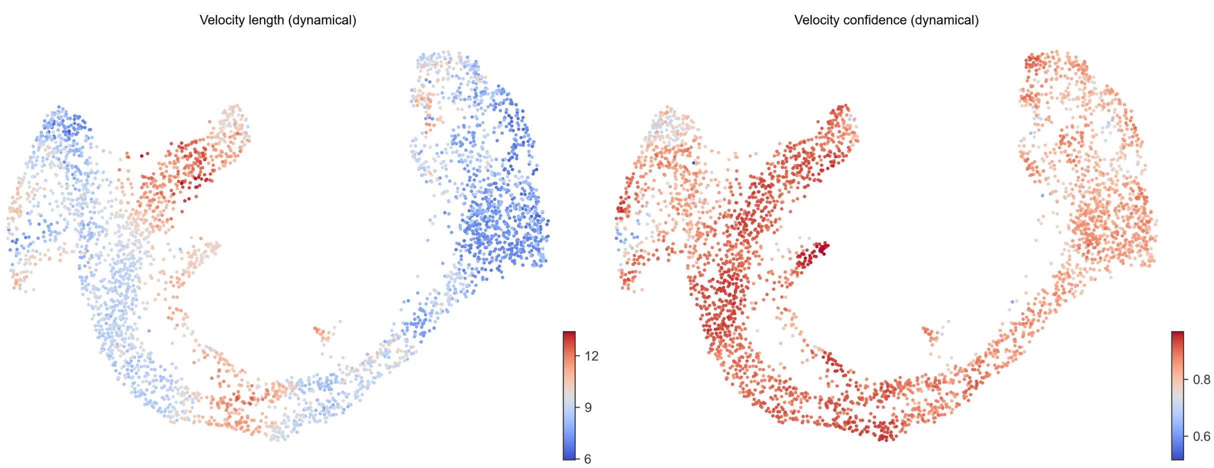

Figure 2: Per-cell velocity length and confidence (dynamical mode). Confidence drops in the Ductal compartment, consistent with mixed stem-cell states.

PAGA velocity graph (Figure 3)

The cluster-level transition graph shows two halves: a linear flow from Ductal through Ngn3 low EP to Pre-endocrine on the right, and a fan-out from Pre-endocrine into the four mature endocrine fates on the left. This is the expected developmental trajectory of the embryonic pancreatic endocrine system.

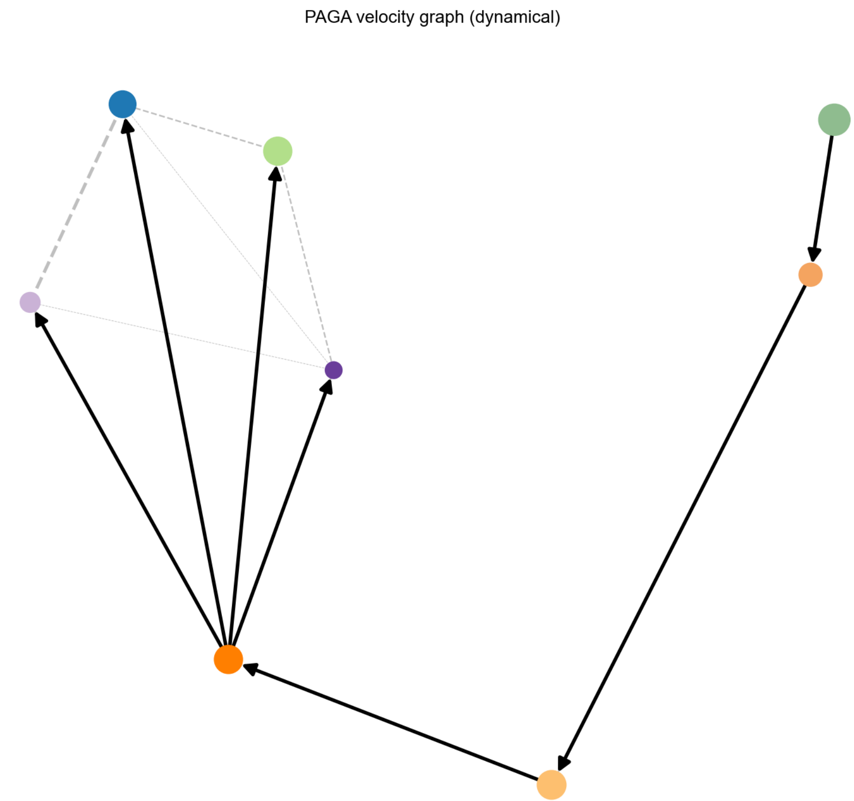

Figure 3: PAGA velocity graph (dynamical). Arrows show inferred directed transitions between annotated clusters.

Latent time and driver genes (Figures 4-5)

The dynamical model's latent time is a gene-shared estimate of how far each cell is along its trajectory. Coloring the UMAP by latent time recovers a clean gradient from Ductal (t = 0, dark) through the progenitor populations to mature endocrine cells (t = 1, bright) - a developmental ordering recovered purely from splicing kinetics, independent of any annotation.

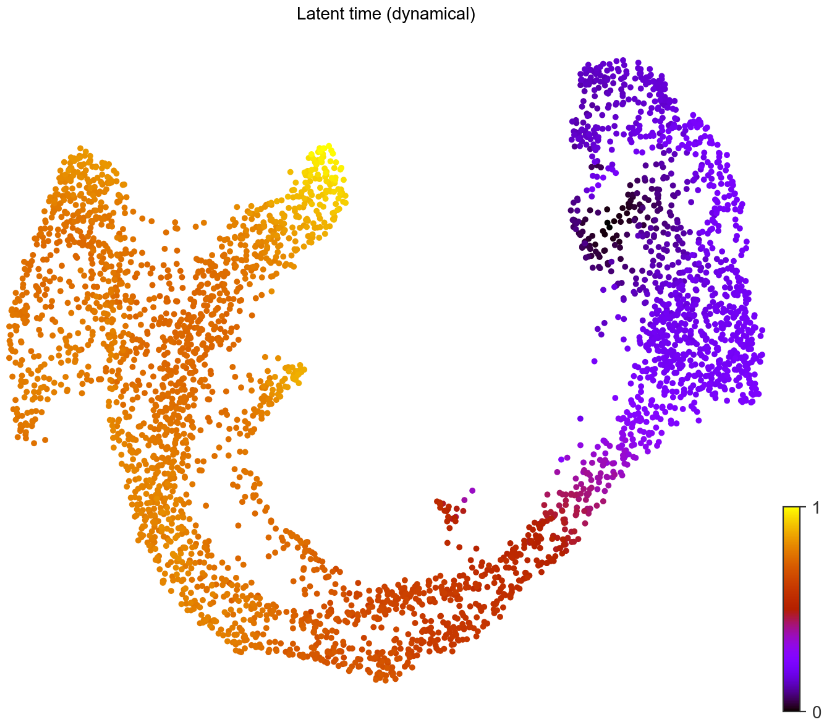

Figure 4: Latent time inferred by the dynamical model. Gradient runs from Ductal (t=0) to mature endocrine fates (t=1).

The top likelihood-ranked driver genes recapitulate the known biology of endocrine commitment. Early in latent time the heatmap is dominated by cell cycle and proliferation genes (Top2a, Mki67, Cdk1, Hmga2, Nusap1, Kif15); the mid-trajectory is enriched for Ngn3-low/pre-endocrine markers (Smoc1, Dcdc2a, Bicc1, Shank2, Nfib); the late phase is dominated by canonical endocrine maturation markers (Pcsk2 - prohormone convertase; Abcc8 - sulfonylurea receptor / K-ATP channel subunit; Pak3, Tmem163, Cpe, Gnao1, Tspan7).

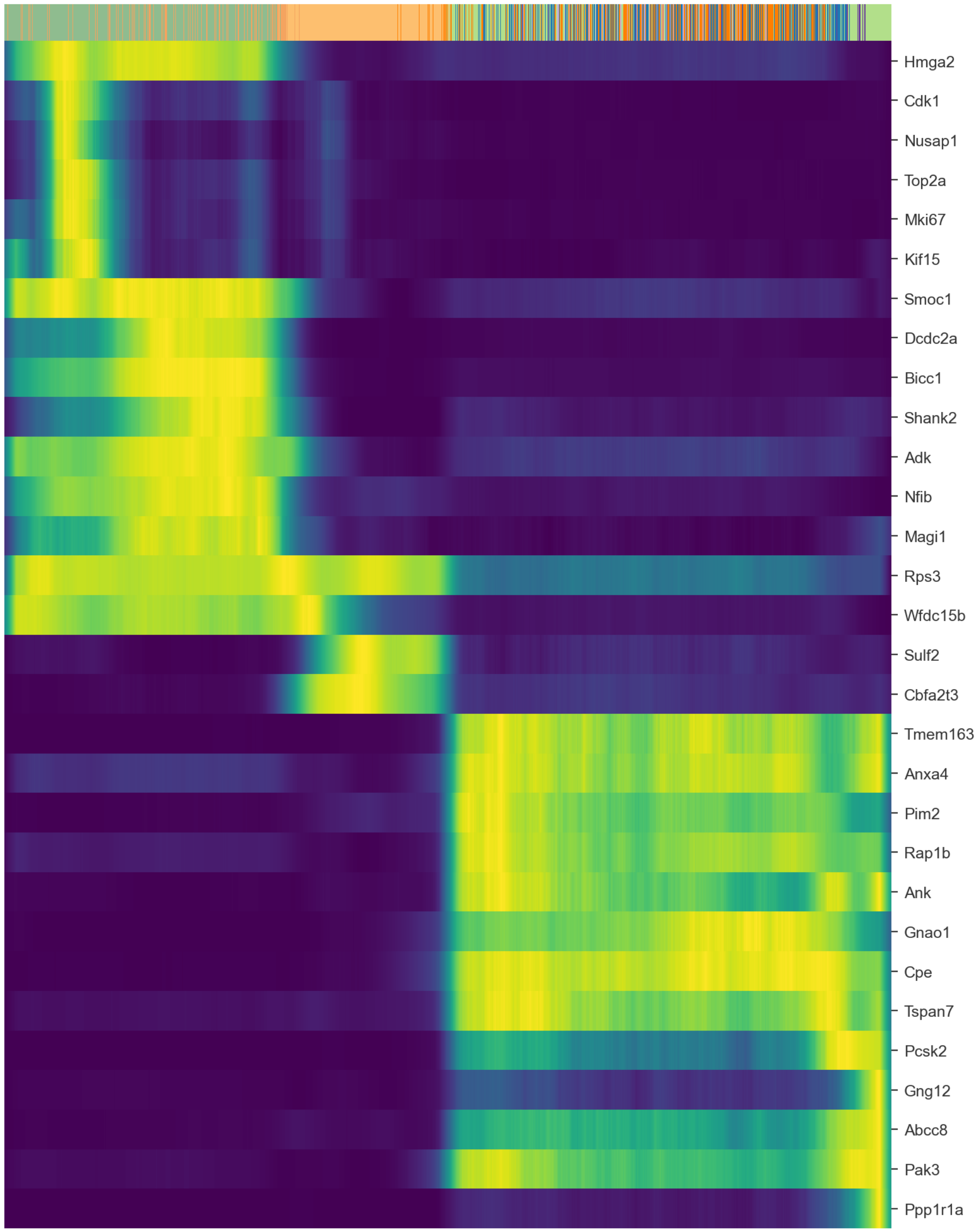

Figure 5: Top 30 driver genes by dynamical-model fit likelihood, ordered along latent time. Early cell-cycle / ductal markers transition through Ngn3-low programs to mature endocrine genes.

Quantitative comparison

Metric	Stochastic	Dynamical
Genes fit (non-NaN velocity)	7,197 (100%)	2,722 (37.8%)
Mean per-cell confidence	0.792	**0.858**
Median per-cell confidence	0.812	0.864

The dynamical model is more conservative - it discards 62% of input genes whose kinetics it cannot reliably fit - but the retained subset produces a higher-confidence velocity field. Per-cell cosine similarity between the two velocity vectors, computed over the 2,722 commonly fit genes, has a global mean of 0.158 (median 0.139). The agreement is non-trivial but modest, with substantial cluster-level structure:

Cluster	Mean cosine	n cells
Beta	**0.283**	591
Pre-endocrine	0.163	592
Ngn3 high EP	0.146	642
Ductal	0.124	916
Ngn3 low EP	0.122	262
Alpha	0.119	481
Delta	0.118	70
Epsilon	0.112	142

Beta cells show the strongest agreement between the two modes, roughly twice the population mean. This is consistent with their mature, highly active secretory-granule transcription program: both models converge when splicing dynamics is strong and unambiguous. In stem-like or transitional populations (Ductal, Ngn3 low EP), the moment-based stochastic model picks up signal from genes that the dynamical model rejects as unfit, producing divergent per-cell vectors despite identical global flow.

Discussion

Mode selection. The two modes are complementary. Use **stochastic** when a fast, robust direction estimate over the entire transcriptome is sufficient - typical for exploratory analysis and large atlases. Use **dynamical** when downstream tasks require per-gene kinetics, latent time, or driver-gene ranking; this is the gold standard for figure-quality velocity in publications.

Biological recovery. The latent time gradient and the top driver gene set agree with the published account of embryonic endocrine commitment: proliferation-driven ductal expansion, Ngn3-mediated lineage specification, and maturation programs centered on Pcsk2 / Abcc8 / Pak3. No additional manual annotation was required to recover this picture - the splicing signal alone ordered the data correctly.

Cluster-level agreement. The Beta-specific increase in stochastic-vs-dynamical cosine agreement is a useful diagnostic: regions with mature, coherent transcriptional programs are robustly fit by both algorithms, while transitional or stem-like populations are inherently more ambiguous. A consultant deploying velocity in a new system should expect similar variance across maturity gradients.

Limitations

- The dataset has a published author annotation; the analysis here re-validates the annotation using independent splicing-based ordering. For a fully blind experiment on a novel system, manual interpretation of latent-time endpoints and driver genes would still be required.
- The dynamical model's NaN rate (62%) is a known characteristic of the likelihood fit; alternative approaches (CellRank2, DeepVelo) reduce this at the cost of additional model complexity.
- All inference here is CPU-only; on this dataset (3,696 cells) runtimes are minutes, but scaling to >100k cells would benefit from a GPU.

Reproducibility

- Source: https://github.com/zivanovicmkg/scrnaseq-portfolio
- Project subdirectory: `06_rna_velocity_pancreas/`

- Environment: scanpy 1.11.5, scvelo 0.3.4, anndata 0.12 (NumPy 1.26 required for scVelo 0.3.4 compatibility).
- Total runtime on a 16-core / 62 GB workstation: ~12 minutes.

References

- Bastidas-Ponce A, et al. *Comprehensive single cell mRNA profiling reveals a detailed roadmap for pancreatic endocrinogenesis.* Development 146, dev173849 (2019).
- Bergen V, Lange M, Peidli S, Wolf FA, Theis FJ. *Generalizing RNA velocity to transient cell states through dynamical modeling.* Nat Biotechnol 38, 1408-1414 (2020).
- La Manno G, et al. *RNA velocity of single cells.* Nature 560, 494-498 (2018).
- Wolf FA, et al. *PAGA: graph abstraction reconciles clustering with trajectory inference through a topology preserving map of single cells.* Genome Biol 20, 59 (2019).